Practice Question

Assignment On Approximation

Directions (Q.1-5): What approximate value will come in place of question mark (?) in the following questions?

(You are not expected to calculate the exact value.)

1. 180 % of 234 + 120% of 632 = ?

1) 1150

2) 1160

3) 1170

4) 1180

5) None of these

2. 190% of 3242 = ?

1) 6110

2) 6160

3) 7110

4) 7180

5) None of these

3. $(105.98)^2 + (111.99)^2 - (98.11)^2 = ?$

1) 14175

2) 15160

3) 16170

4) 17180

5) None of these

4. $108.99 \times 108.111 + 304 \times 14.899 = ?$

1) 16000

2) 16330

3) 16770

4) 17180

5) None of these

5. $62413 \div 15 + 628.463 = ?$

1) 3950

2) 4160

3) 4790

4) 4870

5) None of these

Directions (Q.6-10): What approximate value will come in place of question mark (?) in the following question? (you are not expected to calculate the exact value).

6. $499.99 + 1999 \div 39.99 \times 50.01 = ?$

1)3200

2)2700

3)3000

4)2500

5)2400

7. $[(7.99)^2 - (13.001)^2 + (4.01)^3\ ]^2 = ?$

1) -1800

2) 1450

3)-1660

4)1660

5)-1400

8. $\frac{601}{49} \times \frac{399}{81} \div \frac{29}{201} = ?$

1) 520

2)360

3)460

4)500

5)420

9. $(\sqrt[3]{795657} \times 7) \div (3.8 \times 5.5) = ?$

1) 48

2) 22

3)43

4)26

5) 31

10. $98 \times 785 \div (285)^2 = ?$

1) 0.3

2)1.8

3)2.2

4)0.9

5)0.08

Directions (Q.11-15) : What is the approximate value should come in place of question mark (?) in the following questions? (You are not needed to calculate the exact value)

11. $\sqrt{1567.763} \times \sqrt{4400} \div 326.9827 + 1223.293 = ?$

1)1320

2)1240

3)1230

4)1340

5)1430

12. $\sqrt[3]{?} \times 38.76 + 67.88 \times 18.8565 - 79.834 \times 11.343 = 985.3438$

1)3275

2) 3725

3) 3265

4)3375

5)3075

13 $40.93\sqrt{?} + \sqrt{6625} + \sqrt{6920} + \sqrt{?} = 205.7542 \times 7.654$

1)1225

2)1135

3)2125

4) 1125

5) 1235

14. $\sqrt{1276.373} \times \sqrt{2700} \div 897.2569 + 1915.375 = ?$

1)1820
2)2020
3)2120
4) 1920
5) 1720

15. $\sqrt[3]{?} \times 32.87 + 59.83 \times 28.7665 - 48.8745 \times 21.642 = 1085.344$

1)2100

2)3200

3) 2020

4)2120

5) 2200

Direction (Q.16 - 20): What approximate value should come in place of the question-mark(?) in the following question (You are not expected to calculate the exact value).

16. 90.004% of 9500 + 362 = ?

1) 8890

2) 8910

3) 8930

4) 8950

5) 8980

17. 31% of 3300 +659 = ?

1) 1680

2) 1690

3) 1700

4) 1800

5) 1900

18. 456 x 99.999 + 654 = ?

1) 46254

2) 46264

3) 46274

4) 46284

5) 46294

19. 999.99 + 99.99 + 99= ?

1) 1169

2) 1179

3) 1199

4) 1189

5) 1159

20. $(124.99)^2$ = ?

1) 15593

2) 15603

3) 15613

4) 15623

5) 15633

Directions (Q 21-25): What approximate value should come in place of the question-mark (?) in the following question (You are not expected to calculate the exact value).

21. $-(8.002)^3 + (30.001)^2 - (4.01)^4 = ?$

1)100

2)129

3) 200

4)119

5)139

22. $(91.004)^2 - (40.003)^2 - (52.9)^2 = ?$

1)3558

2)3555

3) 3785

4)3885

5)3789

23. $\sqrt{1297}$ **+ 189.23 =?**

1)215

2)225

3) 235

4)245

5) 255

24. 21% of 6600 + 453.987 =?

1)1899

2)1750

3) 1700

4)1800

5) 1840

25. 49.99% of 5400 + 923=?

1)3456

2)3765

3) 3623

4)2876

5) 3890

Directions(Q26-30) What approximate value should come in place of the Question mark (?) in the given equation.

26. $\frac{\sqrt{845}\times 19.932+\sqrt{4230}\times 14.385}{\sqrt{1765}\times 4.877}$ **= ?**

1) 3

2) 2

3) 4

4) 7

5) None of these

27. 3.55% of 8120 – 66.66% of 540 = ? – 28% of 5500

1) 1520

2) 1380

3) 1460

4) 2460

5) None of these

28. $95\frac{14}{25} + 56\frac{22}{45} - 27\frac{3}{5}$ **= ? + 44.44**

1) 70

2) 80

3) 90

4) 100

5) None of these

29. 157.78% of 4820 + 92.33% of 2840 = ? + 115.55% of 1980

1) 8052

2) 9338

3) 7932

4) 6334

5) None of these

30. $(4.8)^2 + (8.14)^2 - (7.09)^2 + (5.6)^2 + (3.24)^2$

1) 85

2) 95

3) 105

4) 75

5) None of these

Direction (Q31-35) what will come in place of question mark (?) in the following question (you do not have to calculate the exact value)?

31. $\sqrt{3970} \times \sqrt{730} - \sqrt{2400} = ?$

1) 1652

2) 1750

3) 1650

4) 1530

5) None of these

32. $\frac{766}{51} \div \frac{387}{42} \times \frac{121}{13} = ?$

1) 17

2) 15

3) 16

4) 18

5) None of these

33. 2.5% of 842 ÷ 0.07% of 421 =?

1) 70

2) 71

3) 73

4) 75

5) None of these

34. $11\frac{2}{13} + 5\frac{2}{11} - 3\frac{4}{9} = ?$

1) 11

2) 13

3) 14

4) 15

5) None of these

35. $1111.25 \times 9.05 + 2323.23 \times 9.05 - 2121.37 \times 9.05$ =?

1) 11817

2) 11810

3) 11815

4) 11816

5) None of these

Directions (Q36-40): What approximate value should come in place of the question-mark(?) in the following question (You are not expected to calculate the exact value).

36. $[(\sqrt{782} \times 35.003) \div 14.08] \times ? = 5932.13$

1)89

2)91

3)74

4)85

5)70

37. (74.567% of 425) - (69.55% of 820) + (58.95 % of 799) = ?

1)136

2) 140

3)217

4) 252

5) 278

38. 979.001 + 4.0087 $\times$ 82.79 = ?

1)1501

2)1381

3)1421

4)1311

5)1221

39.$(12.073)^2 + (28.003)^2 + (34.98)^2 + (84.96)^2$ = ?

1)8188

2)9378

3) 7848

4) 5898

5) 8828

40. $(\sqrt{3135.65} \times \sqrt{3600}) \div 80.445 + 1182.98 = ?^2$

1)85

2)45

3)35

4)65

5)95

Directions (41-45): What approximate value should come in place of the question-mark(?) in the following question (You are not expected to calculate the exact value).

41. 7652 ÷ 8 + 899.99 – 649.899 + 725.002 = ? + 235.78 + 810.05

1) 785

2) 912

3) 821

4) 920

5) 750

42. 29.99% of 799.99 + 40.001% of 720 = ?% of 599.99

1) 68

2) 47

3) 52

4) 88

5) 72

43. 178.999 – 18.001 + 26.999 + 10.999 = 32.677 + 42.875 + ?

1) 123

2) 105

3) 132

4) 118

5)155

44. $(54.99)^2 \times (13.004)^2 \times (29.57)^2 = ? \times (5.001)^2 \times (11.299)^2 \times (15.299)^2$

1) 35

2) 32

3) 26

4)18

5) 41

45. $(22.999)^3 + (16.999)^3 = ? + (18.999)^3 + (12.999)^3$

1) 8024

2) 7900

3) 7880

4) 8860

5) 8200

Directions (46-50): What approximate value should come in place of the question-mark(?) in the following question (You are not expected to calculate the exact value).

46. 131.06% of 462.57 + 341.002% of 119.95 = 158.89% of ?

1)724

2)718

3)522

4)612

5)638

47. $\sqrt{145.92\ \%\ of\ 2172.05} = 23\frac{1}{3}\ \%$ of ?

1)122

2)316

3)134

4)240

5)255

48. 26596.99 + 28435.228 + 497.99 – 3250 = ?

1)52280

2)50210

3)57210

4)55422

5)58123

49. $\mathbf{162.32001} \times$ **44.999** $\times$ **21.999 = ?**

1) 132660

2) 160380

3) 152760

4) 14860

5) 13960

50. $\sqrt{\mathbf{1124}} \times \sqrt{\mathbf{2404}} \times \sqrt{\mathbf{3847}}$ **= ?**

1)100923

2)103764

3)101529

4) 103292

5)100456

Solution:-

Sol-1. 4; 421.2 + 758.4 = 1179.6

Sol-2. 2; 6159.8

Sol-3. 1; 11236 + 12544 -9604 = 14176

Sol-4. 2; 11772 + 4560 = 16332

Sol-5. 3; 4161 + 629 = 4790

Sol-6. 3

Sol- 7. 4

Sol-8. 5

Sol-9. 5

Sol-10. 4

Sol-11. 3

Sol-12.]4

Sol-13. 1

Sol-14. 4

Sol-15. 5

Sol-16. 2;

Sol-17. 1;

Sol-18. 1;

Sol-19. 3;

Sol-20. 4;

Sol-21. 2

Sol-22. 4

Sol-23. 2

Sol-24. 5

Sol-25. 3

Sol-26. 4

$$\frac{\sqrt{845}\times 19.932+\sqrt{4230}\times 14.385}{\sqrt{1765}\times 4.877} = \frac{\sqrt{841}\times 20+\sqrt{4225}\times 14}{\sqrt{1764}\times 5}$$

$$= \frac{29\times 20+65\times 14}{42\times 5} = \frac{580+910}{210} = \frac{1490}{210} = 7.09 = 7$$

Sol-27. 3

3.55% = 3.55/100 = .0355 = 1/29 (approx)

66.66% = 66.66/100 = 2/3

$$\Rightarrow 8120 \times \frac{1}{29} - 540 \times \frac{2}{3} + \frac{28\times 5500}{100} = 280 - 360 + 1540 = 1460$$

Sol-28. 2

$95\frac{14}{25} + 56\frac{22}{45} - 27\frac{3}{5} = ? + 44.44$

$(95 + 56 - 27) + (\frac{14}{25} + \frac{22}{25} - \frac{3}{5}) = ? + 44$

124 + (0.5 + 0.8 - 0.6) = ? + 44

124 + 0.7 = ? + 44

? = 80 (approx)

Sol-29. 3

158% of 4820 + 92% of 2840 = ? + 116% of 1980

(100% of 4820 + 50% of 4820 + 8% of 4820) + (100% of 2840 – 8% of 2840) = ? + (100% of 1980 + 10% of 1980 + 6% of 1980)

(4820+2410+385.6) +(2840-227.2) = ? +(1980+198+118.8)

7616+2613=?+2297

? = 7932

Sol-30. 1

$5^2 + 8^2 - 7^2 + 6^2 + 3^2 = 25 + 64 - 49 + 36 + 9$

Sol-31. 1

$\sqrt{3969} \times \sqrt{729} - \sqrt{2401}$

$= 63 \times 27 - 49 = 1652$

Sol-32. 2

$\frac{766}{51} \div \frac{387}{42} \times \frac{121}{13}$

$15 \times \frac{1}{9} \times 9 = 15$

Sol-33. 2

$\frac{842 \times 2.5\%}{421 \times 0.07\%} = \frac{21.05}{0.2947} = 71.42 \approx 71$

Sol-34. 2

$\frac{145}{13} + \frac{57}{11} - \frac{31}{9} = 11 + 5 - 3.5 = 12.5 \approx 13$

Sol-35. 1

Solution: $1111.25 \times 9.05 + 2323.23 \times 9.05 - 2121.37 \times 9.05$

$1111 \times 9 + 2323 \times 9 - 2121 \times 9$

$9\,(1111 + 2323 - 2121) = 9 \times 1313 = 11817$

Sol-36. 4

$[(\sqrt{782} \times 35.003) \div 14.08] \times ? = 5932.13$

$[(28 \times 35) \div 14] \times ? = 5932$

$[980 \div 14] \times ? = 5932$

$70 \times ? = 5932$

$? = 84.74 \approx 85$

Sol-37. 3

(74.567% of 425) + (69.55% of 820) + (58.95 % of 799) = ?

(75% of 425)- (70% of 820) + (59 % of 800) = ?

319 -574 + 472 = ?

? = 217

Sol-38. 4

$979.001 + 4.0087 \times 82.79 = ?$

$979 + 4 \times 83 = ?$

979 + 332 = ?

? = 1311

Sol-39. 2

$(12.073)^2 + (28.003)^2 + (34.98)^2 + (84.96)^2 = ?$

$(12)^2 + (28)^2 + (35)^2 + (85)^2 = ?$

144 + 784 + 1225 + 7225 = ?

? = 9378

Sol-40. 3

$(\sqrt{3135.65} \times \sqrt{3600}) \div 80.445 + 1182.98 = ?^2$

$(56 \times 60) \div 80 + 1183 = ?^2$

$3360 \div 80 + 1183 = ?^2$

$42 + 1183 = ?^2$

$?^2 = 1225$

? = 35

Sol-41. 2

$7652 \div 8 + 899.99 - 649.899 + 725.002 = ? + 235.78 + 810.05$

$956 + 900 - 650 + 752 = ? + 236 + 810$

$1958 = ? + 1046$

$? = 912$

Sol-42. 4

29.99% of $799.99 + 40.001\%$ of $720 = ?\%$ of 599.99

30% of $800 + 40\%$ of $720 = = ?\%$ of 600

$240 + 288 = \frac{?}{100} \times 600$

$? = \frac{52800}{600} = 88$

Sol-43. 1

$178.999 - 18.001 + 26.999 + 10.999 = 32.677 + 42.875 + ?$

$179 - 18 + 27 + 11 = 33 + 43 + ?$

$199 = 76 + ?$

$? = 123$

Sol-44. 3

$(54.99)^2 \times (13.004)^2 \times (29.57)^2 = ?^2 \times (5.001)^2 \times (11.299)^2 \times (15.299)^2$

$3025 \times 169 \times 900 = ?^2 \times 25 \times 121 \times 225$

$?^2 = 676$

$? = 26$

Sol-45. 1

$(22.999)^3 + (16.999)^3 = ? + (18.999)^3 + (12.999)^3$

$12167 + 4913 = ? + 6859 + 2197$

$17080 = ? + 9056$

$? = 8024$

Sol-46. 5

131.06% of 462.57 + 341.002% of 119.95 = 158.89% of ?

$\frac{131}{100} \times 462 + \frac{341}{100} \times 120 = \frac{?}{100} \times 159$

$605.22 + 409.2 = \frac{?}{100} \times 159$

$1014.42 = \frac{?}{100} \times 159$

? = 638

Sol-47. 4

$\sqrt{145.92\,\%\, of\, 2172.05} = 23\frac{1}{3}\,\%$ of ?

$\sqrt{\frac{146}{100} \times 2172} = \frac{70}{300} \times ?$

$\sqrt{3171} = \frac{70}{300} \times ?$

$56 = \frac{70}{300} \times ?$

? = 240

Sol-48.1

26596.99 + 28435.228 + 497.99 – 3250 = ?

26597 + 28435 + 498 – 3250 = 52280

Sol-49. 2

$162.32001 \times 44.999 \times 21.999 = ?$

$162 \times 45 \times 22 = 160380$

Sol-50. 4

$\sqrt{1124} \times \sqrt{2404} \times \sqrt{3847} = ?$

$35 \times 49 \times 62 = 103292$